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HEALTH INFORMATION SYSTEM

By

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INTRODUCTION:

The course is designed to equip the health records and information officers' students with the competences that will enable them organize and manage health information system.

Objectives/Course contents:

1. Describe the health information system

2. Describe the expectations or needs for health information

3. Describe the sources of health information

4. Describe the organisation of health information systems in Kenya and flow of information

5. Describe the current national forms or data sets used in collecting data required by the health information services.

6. Organize and manage the procedures for completing and forwarding health information services reports.

7. Organize and manage the collection, analysis, presentation and dissemination of health information.

8. Describe and apply the measures of health indicators; fertility, mortality and others that may affect health care delivery.

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9. Describe sources of vital and health statistics.

10. Describe primary health care concepts in relation to health information.

WHAT IS A HEALTH INFORMATION SYSTEM?

1. Health Information has been defined as information about people's health and how the community, government and non-governmental organisations use the information to plan and provide health care and services.

2. Health information describes the incidence, prevalence and causes of major diseases. It also tells us about the availability and the effectiveness of curative services.

3. It has also been defined as a system that collects and presents health information and is responsible for creation and maintenance of a data base.

4. Another definition of health information is that it refers to organised and interpreted data about health status, resources, activities and events as related to health.

System

Definition: A set of interrelated elements each contributing to the accomplishment of an aggregate. The elements and relationships consist of resources, technologies, activities and actors A health information system does not exist by itself. Health information has been variously described as the "foundation" for better health, as the "glue" holding the health system together, and as the "oil" keeping the health system running (Lippeveld T, 2001). There is however a broad consensus that a strong health information system (HIS) is an integral part of the health system, the operational boundaries of which include: … all resources, organizations and actors that are involved in the regulation, financing, and provision of actions whose primary intent is to protect, promote or improve health (Murray C,

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Frenk J and WHO, 2000). It is universally accepted that health information is essential for health decision-making at all levels of the health pyramid. From the level of individual patient care, to the management of specific health programmes through to the policy level where strategic decisions are made information is an integral part of the health pyramid. The health information system has been aptly described as "an integrated effort to collect, process, report and use health information and knowledge to influence policy-making, programme action and research" (WHO, 2000).

Goal of health information systems

The goal is to develop and maintain an accessible and comprehensive health information system which is flexible to the needs of patients, health care providers, health managers, evaluators and other allied health personnel.

However, what is clear is that health information is much more than the collecting of data. Data have no value in themselves – value and relevance come only when they are analysed, transformed into meaningful information, and used

The objectives of Health Information systems:

• Provide quantitative and qualitative data which is essential for identifying major health problems

• Generate information which can be used to evaluate health policies for planning health programmes and the efficient management of health services

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• Open up a continuous dialogue between those who collect and the users of information with the aim of improving health services

• Disseminate timely information to users, including those who work in research and health administration

• Provide regular feedback to all personnel in the information system which helps to emphasize the important role each of them plays

• Create and maintain a health information data base at district

• Health information as part of PHC is required for health planning, monitoring and Evaluation of health programs

The ultimate objective of a health information system is to produce information for taking action in the health sector. Performance of such a system should therefore be measured not only on the basis of the quality of data produced, but on evidence of the continued use of these data for improving health systems operations and health status (RHINO, 2003).

This principle applies at all levels – at the level of patient care, at the health facility, and at the community, district, national and global levels. A health information system is not a static entity but a process through which health related data are gathered, shared, analysed, and used for decision-making – information is transformed into knowledge for action. These principles also apply to all countries, whatever the level of income and degree of sophistication of the health system. Nor are the boundaries of a health information system confined to health – there is a strong interdependence between health information systems and information systems in other sectors.

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For example, higher levels of female literacy are associated with higher compliance with home treatment for diarrhoea/dehydration in infants; improved sanitation is associated with increased child survival; food and nutrition policies affect the health of children and adults alike. Making links such as these and identifying broad areas of data common to health and other sectors is properly within the responsibility of a health information system.

HISTORICAL DEVELOPMENT OF HEALTH INFORMATION SYSTEM

• In 1970, the MOH became conscious of the need for more useful data to assist in formulation of health policies, the setting of priorities and the evaluation of health care programs

• 1972 - A committee comprising of representatives from CBS, MOH, AG and WHO was set up. Their task was to establish an HIS for Kenya

• A pilot project designed and tested in Mombasa which represented urban districts, Kwale represented peri-urban districts and Kitui represented rural districts

• 1974 – the idea of an HMIS was conceived with the creation of vital statistics unit

• 1975- the creation of an evaluation and research unit

• The project was completed in 1976 with the following two major recommendations

– Merge the section of Vital and Health Statistics of the Division of Communicable Diseases Control and Research and Evaluation section of the Division of Family Health into one entity-The HIS

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– Remove the prefix MED in our data collection and reporting tools and replace it with MOH along the number.

- Med 800 -Out patient Register
- Med 801 -Out patient card
- Med 802 -Out patient tally sheet.
- Med 803 -Immunization tally sheet.
- Med 804 -Monthly Laboratory report
- Med 901 -Discharge sheet.
- Med 902 -Out patient Monthly Activity Report.
- Med 903 -Immunization summary

• In 1978, KMTC Nairobi started a two year professional course on Medical Records Technology which boosted the HIS to a greater extent.

• HIS Department -1980 The functions of this department were:

o Collection , processing , and analyzing of health and management (administrative)data

o Improvement of quality of health data , review and modify where necessary the reporting system to enhance the utilization of data by end users

o Assist in the training of health personnel in medical records keeping and programme evaluation techniques

o Dissemination of health information to end users within the Ministry and other relevant organization

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o HIS was also charged with the task of providing other departments within MOH with the information necessary for planning purposes

o Function as a data bank for various types of data in the Ministry of Health

• In 1982 , another committee of professionals was appointed to review and evaluate the system to establish the accuracy and relevance of the data collected by the system

• In 1984 – in response to the National Policy on District Focus for Rural Development, HIS decentralised is services by establishing offices in all districts

• In the subsequent years, disease specific programs emerged, each with its own HIS, a process that weakened the system and went against the goal. Efforts to integrated the system were initiated to again strengthen the central HIS and hence the integration and revision of data collection and reporting tools. The process has reduced the tools due to identification of core indicators.

• All along the district health reports were manually compiled and posted to the MOH headquarter (HIS) before 15th of the preceding month. Computers by then were dreams to many organisations.

• In 1990, KMTC Nairobi initiated a diploma class in Health Records and information officers basically for in- service officers. The 1st batch of pre- service 3 year course HRIO class commenced the following year.

• In 2008, the department of HIS came up with a reporting system which embraced sending of health care reports electronically. It had series of data sets which were summarized and sent through file transfer protocol (FTP). Every district had its own rights to access the data in FTP though this was purely DHRIO business.

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• In August 2009, the first bachelor degree course was initiated in Kenyatta University.

• In 2010, a new system of reporting came up and has to date been adopted by all districts in Kenya. It actually replaced FTP and has managed to integrate all reporting systems. It borrows from different systems to enhance a strong and comprehensive database for decision making. Constant improvements are still going on to make it even better.

THE NEEDS AND EXPECTATIONS OF HEALTH INFORMATION SYSTEM Mandate of Health Information System

• To collect, process, analyze and disseminate health information to users

Vision of the Health Information System

Be a center of excellent for quality health and health related data and information

Mission of the Health Information System

To provide timely, reliable and accessible quality health information for evidence based decision making to promote the health of the nation

Policy goal

• This policy seeks to enhance availability of comprehensive quality health and health related data and information for evidence based decision making. It seek to address such issues as partnership in data collection and information sharing, guidelines and data processing, and data warehousing as well as instituting standardized mandatory reporting by all care providers and quality in data management in the health sector

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Policy Objectives

a) Promote one health information system in Kenya upon which all shall be committed to.

b) Promote use of health information for evidence based decision making, promote accountability and empower citizens to make healthy choices

c) Promote collection of sufficient, relevant, reliable and quality health statistical data pertaining to the health status of the nation, health services coverage and utilization.

d) Promote and encourage production and dissemination of timely, easily understood health and health related information for evidence based decision making by managers at various managerial levels within the health sector

e) Enhance closer co-operation between producers and users of health related data and information through regular meetings, seminars, training and publications.

f) Promote reporting by all health related statistical constituencies through the use of standardized data collection and reporting tools

Advantages of a functioning health information system:

i. Information is readily available when requested or required, avoiding the last minute rushes to look for data.

ii. Policy makers and managers of programmes use the information for the health and development planning and implementation and are sensitized on importance and effective utilization of information.

iii. Resource mobilization and allocation. Once data is readily available, it is easy to project the required resources by extrapolating from the past trends with a fairly good accuracy. Rule of the thumb should not be the norm or the practice.

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iv. Monitoring trends – Disease surveillance is functional and it is thus easy to predict the occurrence of epidemics of diseases.

v. Problem and gap identification – Various departments and institutions becomes part of the system; there is collaboration and co-ordination among various actors.

vi. The work and duties of the staff that are involved in the system is appreciated and recognized.

vii. Ad-hoc surveys and consultancies are avoided.

viii. Decision making and priority service setting.

ix. Operational research and training becomes easy and enables quick retrieval of the evidences in practice and use evidence based practices for demonstration.

x. Monitoring and evaluation of projects and programmes will be made easier and alongside working corrective measures or change of strategies will be realized in the initial stages.

xi. Donor influence in Health information system is also avoided as in case of reviews.

Health information subsystems

A health information system can be considered to consist of several separate subsystems:

• Disease surveillance and outbreak notification.

• Data generated through household surveys.

• Registration of vital events and censuses (births, deaths and causes of death).

• Data collection based on patient and service records and reporting from community health workers, health workers and health facilities.

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• Programme-specific monitoring and evaluation (for example for TB, HIV/AIDS, and EPI).

• Administration and resource management (including finance/budget, personnel, and supplies).

The function of a health information system is to bring together data from all these different subsystems, to share and disseminate them to the many different audiences for health information, and to ensure that health information is used rationally, effectively and efficiently to improve health action.

A strong health information system is an essential component of sound programme development and implementation, and is a prerequisite for strategic decision-making – ultimately, it provides the basis upon which improved health outcomes depend. In theory, a health information system consists of a process of gathering, sharing, analysing, and using health-related data for decision-making – information transformed into knowledge for action. In this respect the concept of a system as an entity "formed of parts placed together or adjusted into a regular and connected whole" (Chambers Dictionary) is key.

WHO ARE THE MAIN ACTORS IN THE HEALTH INFORMATION SYSTEM?

A health information system should generate information for different uses by different health system actors.

Some of these actors operate at macro-decision level (for example, strategic planning, allocation of resources, and evaluation) while others operate at micro-management level (for

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example, case management, programme management, administration, and deployment of human resources) as shown in figure 4.

The kinds of information required by each type of health system actor differ in ways such as degree of reliability, levels of aggregation, levels of detail, and diversity of topics. Given the range of actors involved and the diversity of potential data items, it is imperative that the health information system has the following interlinked characteristics:

• The ability to identify detailed and disaggregated information items useful for decision-making at various levels within the health care system that are also immediately relevant at the level of data collection.

• The ability to screen and channel to central level only what is most essential and detailed enough for strategic decision-making and policy analysis

Sources of data for HIS

Sources of data may be categorized into two:

Population based data;

• Vital registration

• National population census

• Survey

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Health and disease based data

• Absenteeism records

• Sentinel surveillance

• Routine data collection

• Disease registers

i) Vital registration: eg Birth, death, marriage Certificates etc

This is the data that is derived from vital statistics registries that track all births and deaths as well as certain changes in legal status such as marriage, divorce and migration (registration of place or residence).

In developed countries, who have invested in good registration systems (such as the United States and much of Europe), registry statistics are the best method for estimating the number of births and deaths.

Advantages

- Ensures continuous data collection and thus qualifying possible at any given time
- It is less expensive to manage
- Standardized manner of data collection
- Near complete coverage of vital events

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🎬 Many natality and mortality indicators may be derived from those statistics. Useful in compilation of the denominator eg IMR

Disadvantage

◦ Some of the data variables may be incomplete

ii) Census: Latin word which means “Assess”

A census is referred to as a complete enumeration or count of people in a given country conducted periodically usually every 10 years. A census is usually conducted by a national government and attempts to enumerate every person in a country. However, in contrast to vital statistics data, which are typically collected continuously and summarized on an annual basis, censuses typically, occur after every 10 years or so and thus are not usually the best source of data on births and deaths.

Census does more than just counting people. They typically collect information about families or households, as well as individual characteristics such as age, sex, marital status, literacy/education, employment status and occupation, and geographical location. They also collect data on migration (or place of birth or of previous residence), language, religion, nationality (or ethnicity or race), and citizenship. In countries in which the vital registration system may be incomplete, the censuses are also used as a direct source of information about fertility and mortality;

One of the major limitations regarding census is that 10 years is quite a long duration where a lot of developments will have taken place and some events disregarded during the exercise.

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Others include;

• Lack of data for those unavailable during the exercise

• Takes a long time to analyse and disseminate results

• Most notably data for normadic population unreliable

iii) Demographic data service (Survey):

Data is collected from a sample drawn from a given population. The sample picked should have all characteristics of the population studied and thus representative of such population. In demographic data service there is a single round and multiple round surveys. This method requires a lot of planning and administrative and logistical support. It is used to obtain current information on fertility and mortality and other factors surrounding these events. It is used to gauge performance of certain population. They include nutritional surveys, KDHS etc.

iv) Routine data collection:

This entails sources of data for disease burden specifically for mortality and morbidity. They are useful in monitoring performance of health services, resource allocation, quality of care and organisation of the health services.

These records are not fully utilized as required. They contain data which could be used to calculate rates which may be required for planning purposes.

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Weaknesses in routine reporting:

• Constant staff operation required.

• Excessive inappropriate data types requested

• Incomplete late reporting

• Inadequate use of data locally

• Central analysis of data often delayed

• Central action is irrelevant to the local needs.

Disease registers:

• Prevalence of some disease such as cancer, TB,HIV/AIDS are monitored through the disease registers

• These are limited as they are kept only in specilised facilities capable of diagnosing diseases covered by the register.

Definition of Health Indicator

An indicator is a unit of information measured over time that captures changes in a specific condition. They measure progress, performance or change towards meeting program objective. A Health indicator is a characteristic of an individual, population, or environment which is subject to measurement (directly or indirectly) and can be used to describe one or more aspects of the health of an individual or population (quality, quantity and time).

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Health indicators can be used to define public health problems at a particular point in time, to indicate change over time in the level of the health of a population or individual, to define differences in the health of populations, and to assess the extent to which the objectives of a program are being reached.

Health indicators may include measurements of illness or disease which are more commonly used to measure health outcomes, or positive aspects of health (such as quality of life, life skills, or health expectancy), and of behaviors and actions by individuals which are related to health.

They may also include indicators which measure the social and economic conditions and the physical environment as it relates to health, measures of health literacy and healthy public policy. This latter group of indicators may be used to measure intermediate health outcomes, and health promotion outcomes.

Uses of indicators:

• When measured sequentially over time, they show direction and speed of change

• Assist to motivate and put pressure on policy makers to make changes in policies

• Foster more equitable distribution of resources among and within countries or regions.

Selection criteria of an indicator

Measurable: Must be precisely defined to be measurable Valid: Accurately reflects situation it intends to measure Reliable; produce same results every time it is measured regardless of

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who collects data Reliable # valid: A reliable indicator may produce invalid results Sensitive: Must react to changes on intervention Variable: Must show variation between subject and over time. Helps in showing beneficiaries.

Types of Indicators:

A. Demographic Indicators

1. Total Population: The number of people in the country, county, sub-county or any other administrative boundary.

2. Population Growth Rate: The average annual population growth rate is the rate of natural increase in the population plus the net migration rate.

3. Life Expectancy at Birth: The average number of years that a person at age 0 will live if age-specific death rates remain constant. Life expectancy at birth is highly affected by rates of infant and child mortality.

4. Crude Birth Rate: The number of births per thousand of the population. The product of the number of live births, divided by the midpoint population, multiplied by 1,000

5. Crude Death Rate: The number of deaths per thousand of the population. The product of the number of deaths divided by the midpoint population, multiplied by 1,000

B. Socioeconomic Indicators

6. Adult Literacy Rate: The percentage of people ages 15 and over who can, with understanding, read and writes a short simple statement.

7. Gross Enrolment Rate - Primary School: Total enrolment in primary level of education, regardless of age, expressed as a percentage of the official school-age population

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corresponding to primary level of education in give school-year. GER can be over 100% due to the inclusion of over-aged and under-aged pupils/students because of early or late entrants, and grade repetition.

8. Access to safe water source: Refers to the percentage of the population with access to safe water sources, such as a household connection, public standpipe, protected well or spring, or rainwater collection.

9. Access to Sanitation Facilities: Refers to the percentage of population with access to excreta disposal facilities.

C. Family Planning Indicators

10. Total Fertility Rate: The number of children a woman between ages 15-49 would have during her lifetime if she were to bear children at the currently observed rates.

11. Contraceptive Prevalence Rate: Percentage of women aged 15-49 currently using a method of contraception.

D. Maternal Health Indicators

12. Maternal Mortality Ratio: The estimated number of women who die as a result of pregnancy or childbirth per 100,000 live births.

13. Assisted Delivery by a Health Professional: The percentage of births/deliveries that occur with the assistance of any trained health professional including doctors, nurses and midwives.

E. Child Survival Indicators

14. Infant Mortality Rate: The estimated annual number of deaths of infants under 12 months in a given year per 1,000 live births in that same year.

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15. Under-5 Mortality Rate: Annual number of deaths that occur in children 0-4 years old per 1,000 births.

16. Stunted (height-for-age): Percentage of children under age five whose height-for-age is below minus-two standard deviations from the median of the reference population.

17. Underweight (weight-for-age): Percentage of children under age five whose weight-for-age is below minus - two standard deviations from the median of the reference population.

18. Wasted (weight-for-height): Percentage of children under age five whose weight-for-height is below minus-two standard deviations from the median of the reference population.

19. Measles Vaccination Rate: Percentage of living children who have received one dose of measles vaccine.

F. HIV/AIDS Prevention Indicators

20. HIV Prevalence rate: The estimated number of adults aged 15-49 years living with HIV/AIDS divided by the adult population in the same age group.

G. Infectious Diseases Control Indicators

21. TB Case Detection Rate: Percentage of annual new smear-positive notifications of the estimated annual new smear-positive incidence.

22. TB (DOTS) Treatment Success Rate: The proportion of smear positive patients who were cured plus the proportion who completed treatment.

Terms commonly used in indicator measurement;

Baseline data

1. A set of data collected at the beginning of a study or before intervention has occurred.

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2. Baseline: a reference point used to indicate the initial condition against which future measurements are compared.

Population: denoted (n)

1. The total number of people inhabiting a specific area.

2. The set of individuals, items, or data from which a statistical sample is taken.

3. All the organisms that constitute a specific group or occur in a specified habitat

4. All of the objects or items in a specifically defined area considered as a whole.

Population proportion: the percentage of the population that has the subject characteristics.

Population pyramid: a graphic presentation of the composition of a population with the largest group forming the baseline, the smallest at the apex.

Population at risk: the individuals belonging to a certain group or community who have the potential to contract a medical condition or the population which is composed of animals that are exposed to the pathogenic agent under discussion and is inherently susceptible to it. High or special risk groups are those which have had more than average exposure to the pathogenic agent.

Stable population: a population which has constant mortality and fertility rates, and no migration, therefore a fixed age distribution and constant growth rate.

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Target population: in epidemiological terms the population from which an experimenter wishes to draw an unbiased sample and make inferences about it.

Catchment area: The surrounding area served by an institution, such as a hospital or school.

Producers of health information Health sector

The health sector is the primary supplier of health information. It is responsible for information generated through disease surveillance and response efforts designed to provide early warning of disease outbreaks (such as polio, Ebola, and SARS); to illustrate patterns of chronic disease spread (such as HIV/AIDS or cancer); and to produce information relating individual risk behaviours to health outcomes (for example smoking and cardiovascular diseases).

Data related to the performance of health services, to the management of resources for health, and to the policy and legal framework relevant to health are also largely generated through the health sector. Information about the quantity, distribution, reach and quality of health information and service provision, on the resources needed to provide those services, and on the use of information and services by the population can be generated through routine health management information systems (HMIS).

Within the health sector, there are multiple producers of health information and while the public health authorities may be primary (particularly with regard to information for public

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health policymaking) information from the private for-profit and non-profit sectors is also key to an effective health information system. In many settings, health insurance systems are primary producers of data on patterns of disease and health care use. Employers too are also important potential producers of health information. Other branches of industry, such as the pharmaceutical sector, are both producers and consumers of health information. It is rarely evident that the information that can be derived in these ways is linked to the national health information system.

Other sectors

The health sector is also a consumer of information generated by actors external to health. In many settings, the primary data producer for the ultimate health outcome (mortality) is not the health sector. Instead, data on vital events such as births, deaths and sex and age patterns of mortality, are produced through the census and the civil registration system, often under the overall responsibility of ministries of the interior or planning and national statistics offices. On the other hand, the health sector has to work closely with the civil registration authorities to generate information on patterns of causes of death because attribution of cause of death is generally the responsibility of health care professionals. Health-related information may also be produced by local and municipal authorities.

Information on resources allocated and consumed for health generally derive from departments of finance and planning, and resource flows to the health sector can be extracted from national accounts where these are available.

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Surveys, research, and communities

In some settings departments of national statistics are primary producers of health information derived from household surveys – often the sole source of population-based information on the use of health services or health-related behaviours. Although the increasing frequency of poverty monitoring, household budget surveys and household living standards surveys can provide useful health-related data down to provincial level, they are rarely used by health information systems. Household surveys are of particular importance in settings where vital registration is incomplete (which is the case in most developing countries) and may be the sole source of population-based information on important health outcomes such as infant mortality.

Other important producers of health information are academic and research institutions, often supported by external funding. Researchers may develop new tools and methods for assessing different aspects of health and can play a major role in the evaluation of health interventions.

Frequently neglected sources of health information are communities and advocacy groups. Communities can play a major role in gathering information on disease surveillance, births and deaths, environmental issues, follow-up of patients, patterns of health-seeking behaviour and perceptions of health services. Although the information they produce may be considered non-representative or incomplete, and may be limited to qualitative information rather than quantitative statistics, it should be considered a potentially important component of the health information system. The importance of qualitative information, such as stakeholder opinions and perceptions, should not be neglected in health information systems

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Users of health information

At each level of the health care system, users of health information have differing needs and use information in different ways. At the most basic level of client–health worker interactions, patient records are a vital source of information, whose utility is not confined to the individual level. Record reviews can be used to ascertain the extent of conformity with agreed norms and standards of care. Confidential enquiries and facility-based audits review provider practices in order to determine to what extent care could be improved and the degree to which deaths were avoidable and the potential policy implications of such avoidable factors.

At the facility level, managers need information on patient profiles, patterns of admissions and discharges, length of hospital stay, use of medicines and equipment, deployment of different categories of health care workers and ancillary staff, costs and income. At district level, planners and managers use this information and data on locally relevant population profiles and risk factors in decision-making regarding allocation of resources to different facilities. Within the public health sector, such information is transmitted upwards through district and provincial levels to the national level where basic resource allocation decisions are made. More problematic is the extent of such reporting by the private sector – unless there is a strong regulatory framework within which the private sector operates, it is unlikely that such information will be transmitted to the planning authorities.

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Although the health information generated through the reporting of routine activities by health care facilities and health care workers provides important and useful information on the activities of the health system, this is insufficient for strategic decision-making regarding the allocation of health resources. Decision-makers need information not only about service activities and users of services, but also about those who for whatever reason do not use the services. This information is generally harder to come by than routine service statistics. Health care facilities may undertake special studies of their catchment populations in order to ascertain demand or need for information and services. More often, such information is derived from household surveys in which people are asked direct questions about their perceived need for and use of health care services. The major advantage of using household surveys for such information is that it is possible to obtain socioeconomically stratified information on use of all types of service, including the private sector (modern, private-for-profit, private, non-profit, traditional providers, social marketing outlets, pharmacies, etc.). An important disadvantage, however, is that household surveys are generally undertaken at national level and for reasons of costs, sample sizes are generally insufficient to permit detailed analysis at the district level

When making strategic health sector decisions, national level authorities use health-related information from sources such as routine service statistics, household surveys, vital registration, census, national accounts, and education and employment data (particularly with regard to the production and availability of human resources for health). One visible manifestation of this process is the reporting at national level of progress towards national health-related goals such as reductions in child mortality or reduced disease transmission.

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National authorities also report health-related information to international bodies such as the United Nations. Increasing interest in the performance of national health systems has been stimulated by the goals-oriented international conferences of the 1990s, summarized in the Millennium Development Goals, endorsed by 189 heads of state and government in 2000. The progress made by countries towards these and similar goals is of interest to donor agencies and governments desirous of tracking the extent to which external assistance produces tangible results in terms of improved health.

This rapid overview of the different users of health information at different levels demonstrates three important principles:

- Different types of health-related data are needed at different levels of the health care pyramid – not all items of information need to be reported at every level.
- Those working at the periphery, closest to patient management, need more detailed information on clients seen and services provided than those working at the central level.
- In order to avoid overburdening health care workers at the peripheral level, managers and planners should consider carefully what type of data should be generated at each level, bearing in mind that to the extent possible, those collecting and transmitting data upwards through the system need to understand and appreciate why the data are required.

The organisation of health information systems in Kenya and flow of information

Different countries have different health information systems all based on their needs of those countries. In some settings, a highly centralized, rigid and unresponsive health information

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system collects vast quantities of data, only a small proportion of which is used to manage performance, generally through punitive measures. Elsewhere, health information systems have developed in a chaotic and unplanned way, resulting in duplication and overlap coupled with gaps in information on important health topics. In many developing countries, external donors have introduced partial health information systems designed to meet their own particular needs for accountability – the uncoordinated manner in which such donor-driven information systems have evolved for different programmes resulting in duplication of data collection and data sets.

Too often, government policies are weak and donors tend to dominate – while demanding more

Figure 6 -Typical organizational structure of the health information system

and better data for their own reporting purposes. Such data are seldom relevant or used at the national level. Several donors operate their own monitoring systems because of constraints in the national systems. Health care workers find themselves having to report similar information to several different donors or programmes, though generally in different formats. This results in a lack of clarity and accountability regarding available resources, both human and financial.

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Even within ministries of health, there is poor sharing and coordination of data and information from and between different departments, and duplication of data-generation efforts often due to external pressures.

In recent years there has been a significant increase in the availability of computers for storing, transmitting and presenting health information. However, although computers exist and function well at the district and national levels, they are seldom used systematically to analyse data or for electronic communication. In addition, some countries have adopted technology-driven systems that they cannot afford to maintain and do not have the expertise to manage. One important finding from several studies is that the introduction of technology is not necessarily the answer to a weak health information system. Where the will and capacity to manage paper-based systems exists, the introduction of computers is likely to be well managed and fruitful. In the absence of such a supportive environment, computers alone do not provide the solution.

Current national forms or data sets used in collecting data required by the health information services

Cagories of data collection tools

◦ Individual patient/client cards

◦ Patient listings – Registers

◦ Collation forms(eg tally sheets) to collate – to gather several forms so thet you can have one

◦ Aggregation and Reporting

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1. Patient/client-held

🗣Held by client [mostly] for continuity of care

🗣Used for short periods of time

🗣Usually shifts most of their contents to the register

🗣Examples: Maternal health card or Child health card

Pros:

👄 Opportunity for seeking service outside the issuing facility

👄 Decisions can quickly be made by the receiving service provider on the basis of details already on card

Cons:

👄 Highly prone to loss and damage

👄 Propensity for duplicating data elements onto registers

👄 Recipe for gaps when conducting a detailed clinical review for quality of care

👄 Comprise in data integrity - may be falsely updated

Facility-retained [cards]

👄 Privacy and confidentiality

👄 Longevity of use – demand for a detailed history

👄 When used in conjunction with registers – registers only retain data needed for health unit management and higher demands.

Examples; HIV Care/ART Card, HIV-Exposed Infant Card, OPD/IPD cards

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Pros

- Preserves confidentiality - Preserves content integrity - facility retains responsibility of all the contents on the document

- Can be used to verify/update related records on registers

Cons

- Locks client/patient to one facility

- Missing cards in high-volume facilities

- Consumes storage space with time

2. Registers or Patient Listings

◦ Cross-sectional, longitudinal or both

◦ Cross-sectional

- Patients/clients are listed according to the timing of the visit – as a primary grouping criterion

- Usually more than one entry is made for each patient/client [as many as the visits]

- Useful for short-duration services

- May contain detailed client/patient information

◦ Cross-sectional

◦ Examples

- HTC & Laboratory ,Antenatal, Maternity, Postnatal, Minor Theatre

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Pros:

- Groups patients by date of service provision making page summaries possible
- Recorder doesnt have to flip through old entries to update with current interaction

Cons:

- Redundancy – a lot of blank fields in records as not all columns are updated on each visit
- Wasteful – Similar information is repeated on each visit – name, sex, age, address etc.
- Quality review – difficult to review quality of service and data as one record is spread across many pages or registers.

Longitudinal

- Patient listing is organised according to a specific criteria
- One record (line) for each patient usually retained and updated at specified intervals
- Usually contains data for health unit management
- Examples

◦ Immunisation (Permanent) register

◦ HIV-Exposed Infant

◦ ART Monthly

Pros

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- Provides a continuous record in one place
- Easy to review for quality of care

Cons

- Need for additional tools – demands for accompanying cards and separate collation forms
- Difficult to update on the spot – usually updated after service provision from cards
- Propensity for data gaps – when update after the service

3. Collation Tools

- Tally Sheets

◦ Like activity sheet, contains reportable data only [health unit management]

◦ Count of provider-patient/client interactions through use of tallies

◦ Tallying is done on contact or from registers

◦ Do not include patient identifiers

◦ Examples

- Care and Treatment Tally Sheet

Pros

- Good for single-service or fewer service interactions such as in curative care Cons

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- Unreliable – no record link, hence difficult to audit

When used on similar services in more than one department – creates additional collation tasks and some pages go missing

4. Aggregation Forms

Designated form for each level

- Eg PHC, Hospital, District, etc

Example = Cohort Summary Form

Comprehensive forms

o Same form flows through the hierarchy

o Usually in paper form from facility to district level and may be electronic from district upwards

Why introduction of the current data tools:

The collection, processing, analysis and utilization of accurate data at the facility and district levels are of utmost importance in the provision of quality care. The development of an effective health information system will therefore greatly improve the efficiency of the health services delivery at all levels.

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Health Information and Management Systems are critical as stated in the National Policy Framework paper 1994 – 2010, National Health Sector Strategic Plan I and II 1999 – 2004, 2005 -2010 and the Performance Monitoring and Evaluation (PME) Framework 2005 to 2010 respectively.

To meet the existing greatest challenge, the health sector under the coordination of Ministry of Health HIS division initiated the process to review and integrate data collection (Registers) and reporting tools (summary forms) in April 2006 to meet the current challenges and demand for data. The process took over one year to agree on indicators and variables to be included in the data sets or what should be collected by the registers and reporting summaries.

The revised data collection (registers) tools in 2006/07 are as follows:

DATA COLLECTION TOOLS (SERVICE REGISTERS)

1. MOH 204 A Outpatient Register (under five years);

The register is used for both Male and Female below 5 years (0 – 59 months) who are sick and visit the Health Facility for treatment. The patient/client is registered as an outpatient upon issuance of OPD card. The register also covers the IMCI component and should not have any other register for districts implementing IMCI. The register has various data elements which need to be captured in it. These include the patient name, age, sex, first visit or revisit, diagnosis, treatment given, danger signs and others. This register is supposed to

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feed the MOH 701 A Under five (<5) years Daily outpatient morbidity tally sheet before posting the daily summaries to MOH 705 A Under five (< 5) years Daily outpatient morbidity summary sheet.

2. MOH 204 B Outpatient Register (over 5 years)

Just like the MOH 204 A Outpatient Register (under five years), this the register is used for both Male and Female above 5 years (60+ months) who are sick and therefore seek treatment in the health facility. The patient/client is registered as an outpatient upon issuance of OPD card. The register has various data elements which need to be captured in it. These include the patient name, age, sex, first visit or revisit, diagnosis, treatment given and others. This register is supposed to feed the MOH 701 B Over five (>5) years Daily outpatient morbidity tally sheet before posting the daily summaries to MOH 705 B Over five (>5) years Daily outpatient morbidity summary sheet.

3. MOH 511 Child Welfare Clinic (CWC) Register

The register is used at the MCH for clients/children less than 5 years (0 - 59 Months) who attend Child Welfare Clinic (CWC). This register is kept SEPARATE from Permanent EPI register. This register is kept in MCH clinic basically for growth monitoring of the children until they attain five years. It is a cross sectional type of client listings. The child details are captured in the register upon visiting the health facility. The information include the CWC number which should be used until the age of five years. The child Welfare Clinic number that will be given to New clients only visiting any Health Facility for the FIRST TIME. It is usually given once for the child and it should be used throughout the life time when visiting

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any clinic. Other details include childs name, age, village, cellophone number, weight, height, any abnormality, ITNs issued and others. It feeds it information to the MOH 711A as well as MOH 717 (Monthly Workload for hospitals)

4. MOH 405 Ante natal Clinic (ANC) Register

ANC register is completed the 1st time a pregnant mother attends the MCH clinic and this is usually refered to as 1st ANC visit. Any other subsequent visit is considered as revisit. It is recommended that pregnant mothers should make at least 4 attendances during a specific pregnancy. In other words, ANC number is issued to new or 1st visit clients visiting your health facility for that particular pregnancy. Upon re-attendance,the ANC number is derived from the Maternal and Child Health Booklet/ Card. It gives an account of the voluntary visit that the client makes to the clinic for check up or follow ups. It also captures the EDD, the HIV status, any abnormality detected, the number of ANC visits done etc. It feeds it information to MOH 711A Integrated tool for RH, HIV/AIDS, Malaria, TB, and CHANIS summary as well as MOH 717 Monthly workload for hospitals.

5. MOH 510 Immunisation Register for Children (Permanent EPI Register)

The register is longitudinal type and is only held in facilities which offer immunization services to children less than one year. The serial number is issued to the child upon registration in that facility which acts as an identification number on the first attendance and is facility specific. Usually written serially. 1, 2, 3…. Some of the information which is captured include the serial number, child's number, name of the child, parent's name, cellophone number of the childs guardian,physical address, dates when the child was

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vaccinated as per KEPI schedule. Dates when vaccination was done must be indicated and ticks or checks should not be used in those date cells. This register is complements the MOH 702 Immunisation and Vitamin "A" Tally sheet before posting the daily summaries to MOH 710 Immunisation and Vitamin "A" summary sheet. Children served in this clinic use the mother child booklet issued to their mothers during ANC period.

6. MOH 333 Maternity (Delivery) Register

The register should be used in all maternity units for the purpose of recording the processes and procedures during the child delivery. It is a cross sectional register. A separate row should be used for each baby in case of a multiple birth. It is completed when a mother get admitted in the maternity unit. An in-patient admission number is given to all clients/patients admitted to the maternity unit. Some of the information recorded in those registers include date of admission, name of the mother, maternity inpatient number, Date of birth, mode of delivery, child sex, liter of blood lost, tested for HIV and results, birth notification number, who conducted the delivery, date of discharge and others. It feeds it information to MOH 711A as well as MOH 717 Monthly workload for hospitals.

7. MOH 406 Postnatal Register

This register is 1st completed upon the 1st contact with mother following giving birth or delivery. It records the date the mother reports to your health facility either as a first visit or re-visits (recorded as DD: MM: YY). It is a cross sectional register. It records the date of delivery, place of birth, mode of delivery, any observation done, family planning options chosen etc.

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8. MOH 512 Daily Activity (Family Planning) Register

The register is used to record all contraceptives dispensed to the mother clients whenever they call in for family planning service. It accommodates change of methods among of mothers from one contraceptive to another. A mother who has previously on female condoms may opt to change to Oral pills. It captures the opening balances in all the FP commodities, how much was issued and the remaining balance in each page.

9. MOH 209 Radiology Register

Radiology register is confined to the X-ray department in health facility where radiological examinations are conducted. The X-ray number is issued to all patients receiving service from from radiology department. Type of X-ray, radiographer's name, residence are among the information captured in the register. It is a cross sectional register and an important tool which can be used to solve legal issues pertaining to health facility involved and patient or client. It feeds the MOH 717 (Monthly workload report for hospitals).

10. MOH 240 Laboratory Register

The laboratory register are basically confined in all Laboratory Centres/Health facilities with laboratory services. Upon the clinician requesting a laboratory test for a patient, the laboratory technologist initiates a process by registering the patient, type of specimen requested, specimen results, condition of the specimen and who conducted the test. Just like radiology register,it is a cross sectional register and an important tool which can be used to

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solve legal issues pertaining to health facility involved and patient or client. It feeds the MOH 717 (Monthly workload report for hospitals).

11. MOH 301 Inpatient Register

The inpatient register is placed in inpatient desk for registering the patients recommended fro hospital admission. Unlike the OPD number which changes every calendar year, once the patient is admitted he or she retains the same number throughout his or her life of medical care in your facility. The number is important in retrieving the patient notes in future when patient comes for further treatment or management. Thefinal diagnosis of the patient is posted to the inpatient register as well as in the patient notes.

12. MOH 268 Diagnostic Index Card

This card acts as a collation tool as well as data collection tool. It derives information from the patient notes once the patient is discharged. The same information is relayed to MOH 718 Inpatient morbidity and mortality summary sheet.

DATA REPORTING TOOLS (SUMMARIES)

If the health information system forms are so complicated that those who are to complete them cannot understand them or the forms are not available all the time, data will not be collected properly. Therefore, every effort must be made to simplify forms and to ensure that there is a constant supply of them. Such forms can have a space for some simple on the spot

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analysis e.g. total numbers or percentage to be calculated. The summaries which draw information from the above registers are listed below:

1. MOH 105 Service delivery

2. MOH 701 A Under five (<5) years Daily outpatient morbidity tally sheet.

3. MOH 701 B Over five (>5) years Daily outpatient morbidity tally sheet.

4. MOH 705 A Under five (<5) years Daily outpatient morbidity summary sheet.

5. MOH 705 B Over five (>5) years Daily outpatient morbidity summary sheet.

6. MOH 702 Immunisation and Vitamin "A" Tally sheet

7. MOH 710 Immunisation and Vitamin "A" summary sheet.

8. MOH 704 Child Health and Nutrition Information System tally sheet.

9. MOH 711 A Integrated tool for RH, HIV/AIDS, Malaria, TB, and CHANIS summary.

10. MOH 711 B Integrated tool - RH, HIV/AIDS, Malaria, TB, and CHANIS District summary.

11. MOH 717 Monthly Workload report for hospitals (Service workload for all areas)

12. MOH 268 Diagnostic Disease Index

13. MOH 718 In-patient morbidity and mortality summary sheet.

14. MOH 708 Environmental Health services

Summary:

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Source of data

Collation tool

Aggregation tools

Remarks

- Out Patient card
- MOH 204 A Outpatient register
- MOH 204 B outpatient register
- MOH 701 A <5 years tally sheet.
- MOH 701 B >5 years tally sheet.
- MOH 705 A <5 yrs summary sheet.
- MOH 705 B >5 yrs summary sheet.

Summaries gives the total number of cases seen in OPD on daily basis

- Mother child booklet
- MOH 510 Immunisation

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MOH 702 Immunisation and Vitamin “A” Tally sheet

MOH 710 Immunisation and Vitamin “A” summary sheet.

The easiest way of deriving data is basically from tally

Source of data

Collation tool

Aggregation tools

Remarks

Register

sheet once vaccination is done to a child

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- Mother child booklet
- MOH 405 Antenatal register
- MOH 333 Maternity register
- MOH 406 Postnatal Register

No collation tools but provision of page summaries available in all those registers

MOH 711 A Integrated tool for RH, HIV/AIDS, Malaria, TB, and CHANIS summary.

MOH 711A has different components to be reported on besides maternal safe delivery and ANC services

MOH 512 Daily Activity (Family Planning) Register

No collation tools but provision of page summaries available in all those registers

- MOH 711 A Integrated tool for RH, HIV/AIDS, Malaria, TB, and CHANIS summary.
- Service delivery point FP commodity consumption report

Reproductive health activities reports

- MOH 209 Radiology Register

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- MOH 240 Laboratory Register
- Most of registers outlined above

No specific data collation tools

MOH 717 Monthly Workload report for hospitals

Service workload for all areas

- Inpatient file, MOH 301
- MOH 268 Diagnostic Disease Index

No specific data collation tools

MOH 718 In-patient morbidity and mortality summary sheet.

Coding and indexing of the inpatients conditions involved

Services summaries used as source

No specific data collation tools

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MOH 105 Service delivery report

Derives the data for indicators from all other facility summaries

Standard Operating Procedures (SOPs)

Procedures for completing and forwarding health information reports:

In order to make the data comparable in time and space, data collection is standardized. Sets of forms are designed on which the collected data can be recorded. Some are tally sheets, monthly report forms and annual summary sheets. Each health facility, activities are instantly recorded by tallying or register with or without existing tools and summed up at the end of the day. The sum up of the daily totals makes weekly or monthly totals for each activity available easily. The following sources of error are common and should be avoided:-

a) Forgetting to record in the service/activity register

b) Forgetting to tally.

c) Misclassification

d) Miscalculations and

e) Figure “cooking”

Community level intervention:

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• Record all the services carried out in a family register or record book.

• Extract information from the register/record book to the provided (prescribed) forms.

• Submit the completed form/ record book to your supervisor at the end of each month.

• Update the board displayed in the community with the information from your community unit.

Health facility level intervention:

• Record and maintain all the service registers at service delivery points.

• For immunisation, giving of Vitamin A and growth monitoring services ONLY, record in the immunization register, give the immunization(s) and make a Tally of what has been given before the client leave.

• Using tally sheets provided for other services, extract the information from the register every morning or before closing the previous days business. This will minimize errors, backlogs of work and time for compilation.

• Complete also the daily summaries. This will also assist you in timely completion of the summary form(s) by only making the totals.

• The health facility in-charge should submit all summaries including the community units' reports to the District Medical Officer of Health (DMOH), by 5th of the preceding month.

• File (i.e. permanent attachment of the documents – avoid loose leaves) and maintain a copy of all reports submitted to the next level.

• The health facility should compile service delivery facility indicators, share with the members of the facility management committee. The team should also use their information for daily activities, running of the institution, annual facility planning improve health

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services, request for supplies and monitoring of health services at the community and health facility level.

• Provide regular feedback to the community using organized Chief "Barazas", community health days and other organized community meetings including women group meetings.

District level interventions:

• The district is the foci (first data repository) of the district health data.

• Collect, collate all the summary reporting forms from ALL the available health service institutions in the district.

• Check all reports for errors, omissions, completeness, consistency (VALIDATE) and enter them in the health facility checklist.

• The person receiving the report should give expressed (WRITTEN) feedback summary note to all the reporting facility and remind those who have not reported.

• www.hiskenya.orgOn verifying the data sets from the facilities, data entry ensues through DHIS website. The system has capability of validating the data entered and in case of wrong entry the data is rectified right away before completing the form. The system also allows the user to analyse the data already sent through DHIS. The system can be accessed through . The DHRIO has the "rights" of adding more facilities, adding the users and deactivating users in the DHIS.

• Data entry should be done by 15th of the preceding month.

• Critically analyse in-depth the facility data.

• Discuss the important indicators in the DHMT meetings.

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• Discuss the improvements with the district stakeholders in the District Health Stakeholders Forum (DHSF).

• Share the performance summary monthly with all health facilities, and stakeholders in the district.

• Using available reports plan for integrated targeted supportive supervision to the health facilities/community units.

Provincial level interventions:

- Currently the provincial level does not have an active role in health information compiled at the district level. They only supervise the district performance based on information displayed on DHIS. They no longer compile their districts reports as it is already done in DHIS.

- Share the improvements with the provincial stakeholders.

- Using available reports plan for integrated targeted supportive supervision to the district.

- Provide technical support to the poor performing districts.

National level interventions:

• The server in the ministry of Health may be termed as second data repository.

• The national level rates the district reporting timeliness based on date 15th of the preceding month.

• Critically analyse the district summaries and often give feedback to the districts

• Other sectors or individuals who may need data have limited “rights” to access the data from DHIS website. The provincial heads have limited rights over their districts. They can

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view the data but have no capacity to change it. Others who don't have rights may use the following credentials: User name - guest and password- Guest2011.

• Critically analyse data bearing in mind different users.

• Share the progress report with the heads of the programmes.

• Share the improvements with the other stakeholders.

• Give the PHMT quarterly performance feedback.

Data Validation:

The quality of data depends on;

• Efficiency organisation of the flow of information at the local level.

• Accurate and up-to-date input without undue cost.

• Training of Health personnel at all levels.

• Establishment of data banks

• Enhanced confidentiality.

It is the responsibility of the Health records and information personnel to cross-check the data to determine whether it is accurate, errors or making sense to users before forwarding to the next level. Every month the concerned officers must cross-check data submitted to them and provide feedback to health facilities or any other level. Data corrections must be done by the source. Health statistics suffer from both quantitative and qualitative defects.

These are: -

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• There may be omissions and errors

• Non-response due to

o Negligence of the reporting personnel.

o Communication problem due to inadequate postage fund, poor terrain and road network.

• Lack of adequate cautioning for defaulters.

• Reporting personnel overburdened with other duties.

• Lack of interest

• Insufficient supervision

• Inadequate training of the health personnel at all levels.

Each of the above defects can be strengthened by;

• Carrying out regular supportive supervision.

• Provision of continuous medical education (CME), workshops and seminars.

• Provision of self addressed envelopes or give funds in form of allocations.

• Network with other partners

• Enforce response by prosecuting those not reporting and provide regular feedback on performance.

• Employ more personnel trained in health records and information

• Motivate the staff.

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Data analysis

The analysis of data collected in the health information system is simple and straightforward. Absence of a calculator is not a valid excuse for lack of analysis as analysis can be done without one. The most important data analysis is to estimate coverage for the services offered e.g. what proportion of children less than 1 year complete immunisation schedule before their first birth day? Or what percentage of women delivers without attending antenatal clinics?

Percentage coverage =

Number of cases attended in clinic Total number of cases in catchment population

X 100

It is vital in data analysis to identify appropriate denominators and numerators. Population data are available from the past census and population projections.

Having estimated the catchment population, the next step is to estimate the target population for various services which is the denominator of the coverage fraction. E.g. catchment population = 25,000.

a) Antenatal care

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Target population 4.5 Percentage of 25,000 = (4.5/100) x 25,000 = 1, 125 If number of new visits are 976 therefore coverage = (976/1125) x 100 = 86.76 % If number of re-visits are 2, 090, the mean number of visits per new attendant for antenatal care is new visits + revisits divide by new visits.

= (976+2090) / 976 = 3.1 visits per new antenatal attendant.

b) Morbidity data

Morbidity data are collected in both outpatient and in-patient by age, and diagnosis. Considering new cases gives a picture of the morbidity pattern at the health facility for example;

Top 4 outpatient morbidity cases

Cause

Cases

Percentage

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Total cases Incidence per 1000 population

Malaria

20,540

50.8

822

Diarrhoeal

3,345

8.3

Anaemia

1,553

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3.8

ARI

6,394

15.8

All other diseases

8,566

21.2

Total new cases

40,398

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100

If the catchment population for that facility was 25,000, the incidence rate for malaria would be 20540 * 1000 25,000 = 822 attacks per 1000 per year.

Quality of information:

The quality of information depends on data that is accurate, accessible, complete and timely. Timely data prompts managers to act on time and make planning not to delay. The data collected must be accessible by users to make decisions and carryout proper interventions. Accurate and complete data gives a picture of what is happening in the area in question and the process will: -

a) Provide up to-date information and thereby improve management and evidence based decision making.

b) Relieve supervisors from manual data analysis so that they can spend more time overseeing staff in the field.

c) Make available useful information and provide timely and regular feedback at all levels.

d) Provide more complete information for district, health facility, provincial and national planning i.e.

o Selecting appropriate target groups for various interventions

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o Selecting the most appropriate solution

o Coordinate public and private activities

o Set fees which are affordable

o Set standards for client satisfaction etc.

Presentation of data

In most clinics or health facilities data are gathered on immunisation, Antenatal care, growth monitoring and morbidity every month. The collection of data becomes more meaningful if analysis is done regularly and locally.

Furthermore presentations of data in the form of tables, charts will enhance insight into what is going on in the catchment population with respect to clinic utilization. In statistics, data are classified into qualitative and quantitative data. Qualitative data cannot be counted. A collection of quantitative data in its original form is called raw data.

a) To analyse these data, we obtain a frequency distribution and grouped into categories, age, classes (range of values) or tables.

b) A slight more complicated way of presenting data is through contingency table. A contingency table shows the relationship of two or more variables. E.g. age and sex of patients with blood diarrhoeal or

c) One step further in the analysis is to produce a graph of the data presented in tables and graphs,which provides more insight into frequency distributions and are easier to read.

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d) Data can be summarized by measures of central tendency, mean, mode and median.

e) Data on deaths and births can be done using the mortality and fertility rates.

Data interpretation

Interpretation involves putting the findings of the study/ data into perspective and making appropriate recommendations.

. Challenges in Health Information systems which can affect management decisions

- Problems in processing of data;
- Computer breakdown
- Delay in producing reports
- Volume of data too high for staff use.

Data Quality Audit:

This is conceived as a mean of verifying the reported performance as well as enhancing service monitoring and reporting systems. It reviews the service given, the number of consumers of the service and accuracy of the reporting system.

DOA does not check the quality of services provided but facilitates improvements of the services through usage of the service data generated. Effective programmes with accurate data recording and reporting systems receive the most credit in DQA.

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The goal of the Data Quality Audit (DQA) is to assure that management of services and allocation of resources (funding) are based on sound and accurate data. The specific objectives include;

- Assess the quality of data accuracy, timeliness of service data and reporting system.
- Audit the reported figures given by the institution and estimate the national verification factor (VF), recounted/ reported service data.
- Providing practical feedback to health workers on how to improve the quality of reported data.

What is auditing?

Auditing is essentially examination of book of accounts. For example ideally on EPI the book of accounts are the various data collection tools in use ledger books, bin cards, tally and summary sheets and immunization register.

It is an independent examination and expression of opinion on the financial statements of an entity by an appointed auditor. Without independence it is not auditing.

Auditors examine every document which support any transaction in a business e.g. services offered in the health facilities. Auditing covers behaviours of staff, their professional qualifications and how responsible they are in their duties.

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Auditing exposes those who are defying the standards: ie

- Qualified nurse or health records and information officers who neglect their duties or
- Unqualified staffs dealing with work beyond their scope.
- Staff using KEPI fridge as a table, cupboards, vehicles, computers used for personal use.

Some advantages of Auditing

- Able to detect fraud
- Prevention of errors in future
- Assessment of routine data quality
- Capacity building through findings and recommendations by the audit teams.

What must we do?

- Ensure that all reports are available and complete ie tally sheets, ledger books, summary sheets, bin cards, supervisory books for the audit year and current year.
- Ensure that the data is properly analysed and accurate.
- Proper answers should be available to explain the difference in data submitted since this may affect the verification factor confidence interval.
- Ensure the consistency of the data reported and denominators used.
- Ensure there is a system monitoring completeness and timeliness of reports at district, provincial and national levels.
- Encourage regular written feedback to the health facilities/districts audited.

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- All reports or summary forms should be properly kept and ready for verification.
- The health facilities / districts with missing data in the reports should explain why it is so.
- Ensure the retention of all health records for at least ten years (10) as stipulated in the national policy.

What we must not do?

- Inflate the health facility or district data or reports
- Be fraudulent e.g. same ink, fresh ink
- Missing reports should not be re-written – may result in a "zero score"
- Keep our offices locked or justify by saying the staff with the key is not on duty.
- Fail or sabotage the DQA verification exercise.

Feedback

When giving feedback it is important to emphasize the importance of reporting and outline procedures for proper reporting. The audit team should sit with facility staff and explain the abnormalities detected during the exercise.

Written Feedback:

- Newsletter- gives details of what was observed during the DQA.
- Fact sheets and information summary sheets.
- Published reports
- Public health bulletins

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- Monthly and annual reports
- Newspapers.

DATA SECURITY

Data security is the means of ensuring that (<http://en.wikipedia.org/wiki/Data>)data is kept safe from (<http://en.wikipedia.org/wiki/Data_corruption>)corruption and that access to it is suitably (<http://en.wikipedia.org/wiki/Access_control>)controlled. Thus data security helps to ensure (<http://en.wikipedia.org/wiki/Data_privacy>)privacy. It also helps in protecting personal data.

In simple terms, data security is the practice of keeping data protected from corruption and unauthorized access. The focus behind data security is to ensure privacy while protecting personal or corporate data.

Backup

Def: - Computer security - protection method whereby several duplicate data files are stored on Secondary Storage devices in the event a catastrophic event damages the computer's main file storage system. It is advisable to store backup data files in different locations to guard against loss in the event of a fire, theft, or other unplanned event.

Backup require two types, one to be backup on even days and the other on old day. The storage device required to be stored far from the main storage (server/computer) or operation office. Reports also can be backed-up on institutional email.

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Note: - Backup and Recovery: The goal will be to back up the data from any system on a daily basis. Backup media could be external hard disk, Organizational email account, flash disk, diskette, partioning hard disk so that it will be available in the event of catastrophic failure.

Storage

Def: - The retention of data in any form, usually for the purpose of orderly retrieval and documentation. A device consisting of electronic, electrostatic, electrical, hardware or other elements into which data may be entered, and from which data may be obtained as desired. Storage facilities varies depending on the size of the institution and workload In a facility you require to have a folder or file, shelves, filling cabinet, box file or lockable cupboard to enable you secure the documents. While in a large institution and district you require having a memory stick (Flash disk), Camera and PDAs you require to have a memory card, external hard drive, RW-CD, R-CD, cabinet, shelve and lockable cupboard. The storage device/documents required to be stored far from the main office storage especially sensitive/vital documents.

Network Operations

Take all necessary precautions to prevent any destructive or malicious program (virus) from being introduced to the system. Employ appropriate measures to detect virus infection and employ all appropriate resources to efficiently disinfect any affected systems as quickly as possible.

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Computer Virus Prevention, Detection, and Disinfection:

The goal of the system will be to maintain updated virus protection from a reputable source. Any and all viruses found will be quarantined or the virus will be deleted. Every organization is required to run and maintain their own anti-virus software from an approved source on all computers that have access to the HMIS system.

Strong User Authentication

Authentication is another part of data security that we encounter with everyday computer usage. Just think about when you log into your email or blog account. That single sign-on process is a form authentication that allows you to log into applications, files, folders and even an entire computer system. Once logged in, you have various given privileges until logging out. Some systems will cancel a session if your machine has been idle for a certain amount of time, requiring that you prove authentication once again to re-enter.

Secure access to physical areas containing equipment, data, and software. Strictly safeguard all data including client-identifying information in accordance with the latest technology available and securely protect it to the maximum extent possible. Maintain and administer central and backup server operations including security procedures and maintain backups of the system to prevent the loss of data. Monitor access to all systems that could potentially reveal a violation of information security protocols. Maintain and audit accurate logs of all changes made to the information contained within the database.

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Administrators may request and receive HMIS passwords and User IDs from Central level. Periodically change of passwords for security purposes. Do not release data to any person, agency, or organization without the client's authorization and following the procedures for the releasing of the data. Database at all level should not be handled by many people and should be limited to few persons who would assist in improving security and management of database.

In the event of departments using files or folders, data security is paramount and the documents must be filed in a permanent building where fire extinguishers are accessible, exit doors, big bill board showing "No smoking", filing rooms should be well ventilated, filing equipment should be raised from the ground level in case of water flooding the room .Doors and windows should be strong enough to prevent un authorized entry.

All patients/clients records and information MUST not be access by un-authorized (persons not directly handling the patients/clients

Limit HMIS access to authorized users and follow all procedures in monitoring those who borrow patient's notes from the department.

Preventive maintenance

Any machines, devices or equipment require regular preventive maintenance to improve efficiency and durability.

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• Files/folders require:-

o dusting, and regular filing of loose notes o weeding of inactive notes/records

o filing area should be apportioned into small areas to facilitate or support files to stand upright.

o keep the filing area dry and cool.

• CDS, Flash disk, Memory cards, diskette

o Place on a rack or album

o Keep them in a dry and cool place

o They are fragile and therefore require careful handling.

o Observe proper removal and inserting the device into the computer.

• Computers, laptop, printers, PDAs, Cameras, Duplo machines

o All these equipment are fragile so handle them with utmost care

o Read manufacturers instructions while installing and assembling the machine.

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o Keep them in a dry and cool place

o Place them on firm workstation, raised from ground level.

o Always wipe the equipment and workstations with dry clean wiper.

o Once a quarterly or six months do major dusting using a dust blower.

o Avoid opening of equipments regularly, but it requires a trained technician

o Avoid using oily and wet hands on a keyboard and monitor

o Avoid taking tea or water near the computer keyboard, it may spill over it.

o Follow the right procedures of warm booting and shutting down the devices/machines as required.

o Ensure there is a proper connection of computer cables.

Quality Health Data

Introduction

Quality data is critical to assessing the global burden of disease and developing public health initiatives. We live in an era of unprecedented technological advancement, which has provided us with increased access to data. However, just because data has become more available does not mean that all data is accurate and reliable. This topic describes quality data, identifies how one can ensure the quality of their data, and discusses the challenges associated with obtaining quality data in health care settings.

What is Quality Data?

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Data quality is defined as "the totality of features and characteristics of a data set that bear on its ability to satisfy the needs that result from the intended use of the data."

High quality data effectively satisfies its intended use in decision making and planning. The following are attributes of good quality data organized into fifteen different categories.

1. Access Security: Data must be restricted and kept secure to ensure confidentiality and the protection of civil liberties.

2. Accessibility: Data must be available or easily retrievable.

3. Accuracy: Data must be correct and free of errors.

4. Appropriate Amount of Data: The quantity of data must be appropriate.

5. Believability: Data must be regarded as true and credible.

6. Completeness: Data must be sufficient in breadth, depth, and scope for its desired use.

7. Concise Representation: Data must be represented without being overwhelming.

8. Ease of Understanding: Data must be clear.

9. Interpretability: Data must be in appropriate language and units.

10. Objectivity: Data must be unbiased.

11. Relevancy: Data must be applicable to the task at hand.

12. Representational Consistency: Data must be presented in a consistent format.

13. Reputation: Data must come from a trusted source.

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14. Timeliness: Data should be recorded as quickly as possible and used within a reasonable time period.

15. Value-Added: Data must provide valuable insight.

Ensuring Quality Data

To ensure quality data, the data must be managed correctly from the time of collection until the time of analysis. The data first must be recorded properly on the desired data collection tools. The data to be collected must be well understood by all the persons handling it through trainings and sensitizations and particularly understanding how the data tools are used.

Next, the data needs to be verified and analyzed. Data analysis should be conducted by trained personnel and is normally done by inputting the data into an electronic database

Data should also be cleaned to remove impurities e.g. outliers, wrong data, errors etc

Why Quality Data is Important

-Poor and inaccurate health information hamper the efforts to improve the health of populations. Good public health decision making is dependent on accurate and timely statistics and data. It is critical that quality health data be obtained in order to assess the magnitude and distribution of the disease burden, so that programs can be developed to address health needs worldwide.

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-Vital statistics, such as births, deaths, and causes of death are also critical for addressing health needs and recording progress towards the Millennium Development Goals and other developmental objectives.

-In a clinical setting, quality data is important because it can improve the care provided. Inaculate data can mislead programming or planning purposes since the associated medical errors can lead to long term damage or death in patients, as well as economic losses.

Accurate and complete data gives a picture of what is happening in the area in question and the process will: -

a) Provide up to-date information and thereby improve management and evidence based decision making.

b) Relieve supervisors from manual data analysis so that they can spend more time overseeing staff in the field.

c) Make available useful information and provide timely and regular feedback at all levels.

d) Provide more complete information for district, health facility, provincial and national planning i.e.

-Selecting appropriate target groups for various interventions

-Selecting the most appropriate solution

-Coordinate public and private activities

-Set fees which are affordable

-Set standards for client satisfaction etc.

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Challenges with Obtaining Quality Data in a health care Settings

- Use of un-standardized data management tools
- Lack of trainings/sensitization on use of data tools
- Problems in processing of data;
- Computer breakdown
- Lack of trained staff in data management
- Volume of data too high for staff use.
- Wrong attitude
- Too many tools to fill

DATA WAREHOUSING

A data warehouse is a database used for reporting and data analysis. It is a central repository of data which is created by integrating data from one or more disparate sources. Data warehouses store current as well as historical data and are used for creating trending reports for senior management reporting such as annual and quarterly comparisons.

The data stored in the warehouse are uploaded from the operational systems .

Data warehousing is the recording and storing of healthcare information. A storage device can hold information, process information or both. Devices that process data (data storage equipment) can either access a separate portable recording medium or a permanent component to store and retrieve information.

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Electronic data storage requires electrical power to store and retrieve data. Electromagnetic data may be stored in either an analog or digital format on a variety of media. This type of data is considered to be electronically encoded data, and most electronically processed data storage media are considered permanent storage, or data that will remain stored when power is removed from the device.

Top Five Benefits of a Data Warehouse

The Data Warehouse InstituteAccording to , a data warehouse is the foundation for a successful BI program. The concept of data warehousing is pretty easy to understand—to create a central location and permanent storage space for the various data sources needed to support a company's analysis, reporting and other BI functions.

And it's really important for your business.

BI-Insider.comBut a data warehouse also costs money — big money. The problem is when big money is involved it's tough to justify spending it on any project, especially when you can't really quantify the benefits upfront. When it comes to a data warehouse, it's not easy to know what the benefits are until it's up and running. According to , here are the key benefits of a data warehouse once it's launched.

A Data Warehouse Delivers Enhanced Business Intelligence

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By providing data from various sources, managers and executives will no longer need to make business decisions based on limited data or their gut. In addition, “data warehouses and related BI can be applied directly to business processes including marketing segmentation, inventory management, financial management, and sales.”

A Data Warehouse Saves Time

Since business users can quickly access critical data from a number of sources—all in one place— they can rapidly make informed decisions on key initiatives. They won’t waste precious time retrieving data from multiple sources.

Not only that but the business execs can query the data themselves with little or no support from IT—saving more time and more money. That means the business users won’t have to wait until IT gets around to generating the reports, and those hardworking folks in IT can do what they do best —keep the business running.

A Data Warehouse Enhances Data Quality and Consistency

A data warehouse implementation includes the conversion of data from numerous source systems into a common format. Since each data from the various departments is standardized, each department will produce results that are in line with all the other departments. So you can have more confidence in the accuracy of your data. And accurate data is the basis for strong business decisions.

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A Data Warehouse Provides Historical Intelligence

A data warehouse stores large amounts of historical data so you can analyze different time periods and trends in order to make future predictions. Such data typically cannot be stored in a transactional database or used to generate reports from a transactional system.

A Data Warehouse Generates a High ROI

generated more revenueFinally, the piece de resistance—return on investment. Companies that have implemented data warehouses and complementary BI systems have and saved more money than companies that haven't invested in BI systems and data warehouses.

And that should be reason enough for senior management to jump on the data warehouse bandwagon.

The Data Warehousing Information Center

LGI Systems Incorporated brought to you by

Essay List

- Getting Started by Learning about Data Warehousing
- A Definition of Data Warehousing
- A Definition of Decision Support/Business Intelligence

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- The Case for Data Warehousing
- The Case Against Data Warehousing
- Actions for Data Warehouse Success
- Data Warehousing Gotchas
- Performing Data Warehousing Software Evaluations
- What Data Errors You May Find When Building a Data Warehouse
- Data Warehousing Political Issues
- Aspects of Data Warehouse Architecture
- What to Learn About in Order to Speed Up Data Warehouse Querying
- What to Learn About in Order to Speed Up Data Warehouse Loading
- How to Save Money on Your Data Warehousing Efforts
- Using Data Warehousing in Strategic Decision Making
- Maintenance Issues for Data Warehousing Systems
- What Business Intelligence Tools are Used For
- Are Web Data Analytics Different?
- Favorite Sites For Keeping Current on Data Warehousing
- About the Author of this Site

The Case for Data Warehousing

The following is a list of the basic reasons why organizations implement data warehousing. This list was put together because too much of the data warehousing literature confuses "next

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order" benefits with these basic reasons. For example, spend a little time reading data warehouse trade material and you will read about using a data warehouse to "convert data into business intelligence", "make management decision making based on facts not intuition", "get closer to the customers", and the seemingly ubiquitously used phrase "gain competitive advantage". In probably 99% of the data warehousing implementations, data warehousing is only one step out of many in the long road toward the ultimate goal of accomplishing these highfalutin objectives.

The basic reasons organizations implement data warehouses are:

To perform server/disk bound tasks associated with querying and reporting on servers/disks not used by transaction processing systems

Most firms want to set up transaction processing systems so there is a high probability that transactions will be completed in what is judged to be an acceptable amount of time. Reports and queries, which can require a much greater range of limited server/disk resources than transaction processing, run on the servers/disks used by transaction processing systems can lower the probability that transactions complete in an acceptable amount of time. Or, running queries and reports, with their variable resource requirements, on the servers/disks used by transaction processing systems can make it quite complex to manage servers/disks so there is a high enough probability that acceptable response time can be achieved. Firms therefore may find that the least expensive and/or most organizationally expeditious way to obtain high probability of acceptable transaction processing response time is to implement a data warehousing architecture that uses separate servers/disks for some querying and reporting.

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To use data models and/or server technologies that speed up querying and reporting and that are not appropriate for transaction processing

There are ways of modeling data that usually speed up querying and reporting (e.g., a star schema) and may not be appropriate for transaction processing because the modeling technique will slow down and complicate transaction processing. Also, there are server technologies that that may speed up query and reporting processing but may slow down transaction processing (e.g., bit-mapped indexing) and server technologies that may speed up transaction processing but slow down query and report processing (e.g., technology for transaction recovery.) – Do note that whether and by how much a modeling technique or server technology is a help or hindrance to querying/reporting and transaction processing varies across vendors' products and according to the situation in which the technique or technology is used.

Often a data warehouse can be set up so that simpler queries and reports can be written by less technically knowledgeable personnel. Nevertheless, less technically knowledgeable personnel often "hit a complexity wall" and need IS help. IS, however, may also be able to more quickly write and maintain queries and reports written against data warehouse data. It should be noted, however, that much of the improved IS productivity probably comes from the lack of bureaucracy usually associated with establishing reports and queries in the data warehouse.

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To provide a repository of "cleaned up" transaction processing systems data that can be reported against and that does not necessarily require fixing the transaction processing systems

what data errors you may find when building a data warehouse Please read my essay on for an explanation of the type of "errors" that need cleaning up. The data warehouse provides an opportunity to clean up the data without changing the transaction processing systems. Note, however, that some data warehousing implementations provide a means to capture corrections made to the data warehouse data and feed the corrections back into transaction processing systems. Sometimes it makes more sense to handle corrections this way than to apply changes directly to the transaction processing system.

To make it easier, on a regular basis, to query and report data from multiple transaction processing systems and/or from external data sources and/or from data that must be stored for query/report purposes only

For a long time firms that need reports with data from multiple systems have been writing data extracts and then running sort/merge logic to combine the extracted data and then running reports against the sort/merged data. In many cases this is a perfectly adequate strategy. However, if a company has large amounts of data that need to be sort/merged frequently, if data purged from transaction processing systems needs to be reported upon, and most importantly, if the data need to be "cleaned", data warehousing may be appropriate.

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To provide a repository of transaction processing system data that contains data from a longer span of time than can efficiently be held in a transaction processing system and/or to be able to generate reports "as was" as of a previous point in time

Older data are often purged from transaction processing systems so the expected response time can be better controlled. For querying and reporting, this purged data and the current data may be stored in the data warehouse where there presumably is less of a need to control expected response time or the expected response time is at a much higher level. – As for "as was" reporting, some times it is difficult, if not impossible, to generate a report based on some characteristic at a previous point in time. For example, if you want a report of the salaries of employees at grade Level 3 as of the beginning of each month in 1997, you may not be able to do this because you only have a record of current employee grade level. To be able to handle this type of reporting problem, firms may implement data warehouses that handle what is called the "slowly changing dimension" issue.

To prevent persons who only need to query and report transaction processing system data from having any access whatsoever to transaction processing system databases and logic used to maintain those databases

The concern here is security. For example, data warehousing may be interesting to firms that want to allow report and querying only over the Internet.

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Some firms implement data warehousing for all the reasons cited. Some firm implement data warehousing for only one of the reasons cited.

By the way, I am not saying that a data warehouse has no "business" objectives. (I grit my teeth when I say that because I am not one to assume that an IT objective is not a business objective. We IT people are businesspeople too.) I do believe that the achievement of a "business" objective for a data warehouse necessarily comes about because of the achievement of one or many of the above objectives.

If you examine the list you may be struck that need for data warehousing is mainly caused by the limitations of transaction processing systems. These limitations of transaction processing systems are not, however, inherent. That is, the limitations will not be in every implementation of a transaction processing system. Also, the limitations of transaction processing systems will vary in how crippling they are.

Finally, to repeat the point I made initially, a firm that expects to get business intelligence, better decision making, closeness to its customers, and competitive advantage simply by plopping down a data warehouse is in for a surprise. Obtaining these next order benefits requires firms to figure out, usually by trial and error, how to change business practices to best use the data warehouse and then to change their business practices. And that can be harder than implementing a data warehouse.

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